Comparative G-Factor Analysis of MS-SENSE and MS-CAIPIRINHA in Parallel MRI

Federica Maria Olivotto

This study examines and compares the performance of two parallel imaging techniques, SENSE (Sensitivity Encoding) and CAIPIRINHA (Controlled Aliasing in Parallel Imaging Results in Higher Acceleration). Using MATLAB simulations, it focuses on analysing the effects of coil sensitivity profiles (uniform vs. exponential) and the variation in coil sensitivity between different slices on the g-factor, a key metric that quantifies noise amplification during image reconstruction.

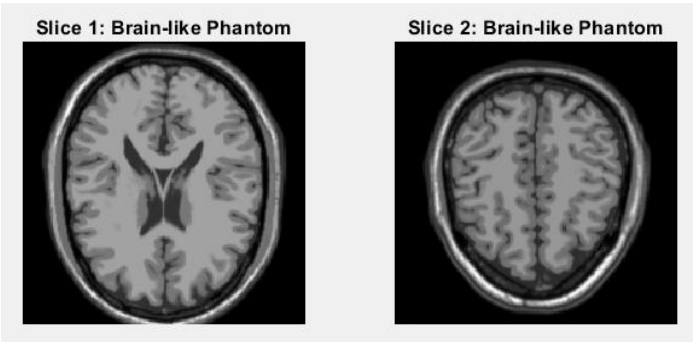
Key points of the Matlab code:

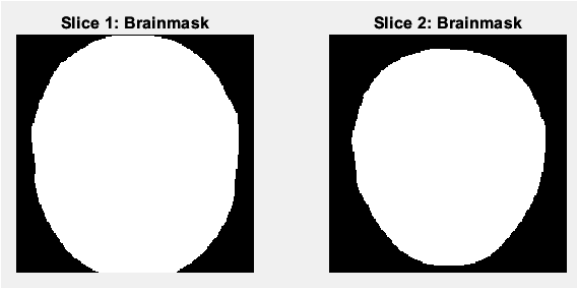
Brain Slices: the brain slices were extracted from a 3D T1-weighted MRI dataset (file name: *T1_ICBM_normal_1mm_pn0_rf0.mnc*), representing a standard "normal" brain anatomy. Each voxel in the dataset corresponds to a 1x1x1 mm cube. In this study, two slices were selected for analysis: the first at the midline of the brain (z = 0), and the second positioned 30 mm above it (controlled by the parameter distance_each_slice).

Noise: since the dataset is inherently noise-free (pn0), we introduced random, uncorrelated noise into the image space, to simulate a realistic acquisition scenario. This noise predominantly originates from the body, under the assumption that the coils have high-quality electronic components. The noise level, controlled by the parameter noise_level, was set to 10 for all simulations. Adding noise was essential to observe the advantages of CAIPIRINHA over SENSE, as noise-free conditions yielded perfect reconstructions for both methods.

Coils: a simplified two-coil configuration was used, with the coils positioned on different planes (separate z-coordinates). Coil 1 was more sensitive to Slice 1, while Coil 2 was more sensitive to Slice 2. The difference in coil sensitivities between the slices was adjustable via the parameter slice_modulation (slice_modulation = 1 indicates identical sensitivities for both slices). For simplicity, it was assumed that coil sensitivities were known, consistent with the SENSE method, where an additional acquisition is performed to measure them.

Binary masks: Binary masks were applied to the coil sensitivities, to focus on specific regions of interest and improve simulation accuracy.





Phase-cycling in k-space: phase-cycling was applied in the CAIPIRINHA implementation, to introduce a shift in image space for the second slice, enhancing the separation of overlapping signals during image reconstruction. This process is governed by the Fourier shift theorem:

$$\rho(x) = \sum_{m=-N/2}^{\frac{N}{2}-1} (S_1(m\Delta k) + S_2(m\Delta k) \cdot e^{-im\Delta k\Delta x}) e^{im\Delta kx} = \rho_1(x) + \rho_2(x - \Delta x)$$

Where:

- $\rho(x) = \rho_1(x) + \rho_2(x - \Delta x)$: final image-space signal, given by the sum of the contributions of the two slices, with the second slice shifted horizontally by: $\Delta x = \frac{FOV}{Nslices} = \frac{FOV}{2}$.
- $S_1(m\Delta k), S_2(m\Delta k)$: k-space signals from the two slices, where m is the phase-encoding step.
- The exponential term $e^{-im\Delta k\Delta x}$introduces the phase modulation needed for the shift.
- $\Delta k = \frac{2\pi}{FOV}$ is the spacing between k-space points.

The transformations from image-space to k-space and viceversa were implemented respectively with the direct and inverse 2D Fast Fourier Transform (fft2 and ifft2); moreover, the functions fftshift and ifftshift were applied in order to centre the zero-frequency component in the frequency domain.

SENSE and CAIPIRINHA reconstruction algorithms:

$$\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \end{bmatrix} = \begin{bmatrix} C_{11}(x,y) & C_{12}(x + FOV/2, y) \\ C_{21}(x,y) & C_{22}(x + FOV/2, y) \end{bmatrix} \cdot \begin{bmatrix} \rho_1(x,y) \\ \rho_2(x + FOV/2, y) \end{bmatrix}$$

The reconstructed images ρ_1 and ρ_2 are obtained by solving the system of equations given by the product between the inverse of the coil sensitivity matrix and the vector of aliased images $[I_1; I_2]$. This is the most critical step of the reconstruction, since in case of similar coil sensitivities in the two slices, the matrix could be nearly singular or non-invertible. In the CAIPIRINHA technique, an additional phase shift is applied to the coil sensitivities for the second slice, aligning with the phase shift performed on the slice itself in k-space. This adjustment helps improve the invertibility of the coil sensitivity matrix.

In MATLAB, this operation was performed pixel-by-pixel, using the pseudoinverse of the coil sensitivity matrix, which is more numerically stable than the standard matrix inverse, especially in cases where the matrix is close to singular or poorly conditioned

Evaluation of results with the g-factor:

The **g-factor** quantifies the noise amplification introduced by the reconstruction process in parallel imaging, when multiple coils are used to separate aliased signals. It indicates the level of difficulty in inverting the coil sensitivity matrix C. It was computed pixel-by-pixel with the following formula:

$$g_p = \sqrt{\frac{Noise\ amplification}{Signal\ sensitivity}} = \sqrt{[(C^H \psi^{-1} C)^{-1}]_{p,p} [C^H \psi^{-1} C]_{p,p}}$$

Ψ represents the noise covariance matrix between the coils. In our case the noise in uncorrelated (white noise), so this parameter corresponds to the identity matrix and can be omitted in the formula. The g-factor doesn't depend on the noise level when the noise is uncorrelated.

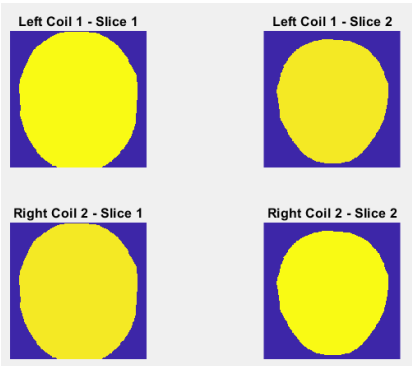
The **g-factor** is inversely proportional to the signal-to-noise ratio (SNR). Ideally, the g-factor should be 1, which corresponds to no noise amplification during reconstruction. As the g-factor increases, the SNR decreases, making the image noisier.

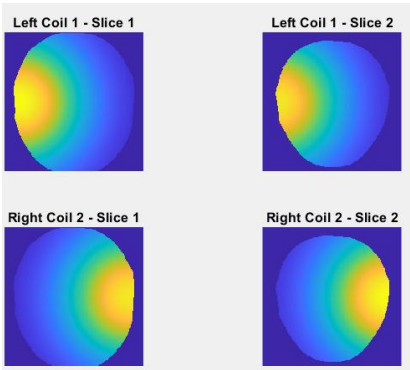
$$SNR_{SENSE} = \frac{SNR_{full}}{g\sqrt{R}}; \quad SNR_{CAIPI} = \sqrt{Nslices}\frac{SNR_{full}}{g\sqrt{R}}$$

R is the in-plane acceleration factor (equal to 1 in our case) and SNR_{full} is the SNR obtained without parallel imaging.

Results

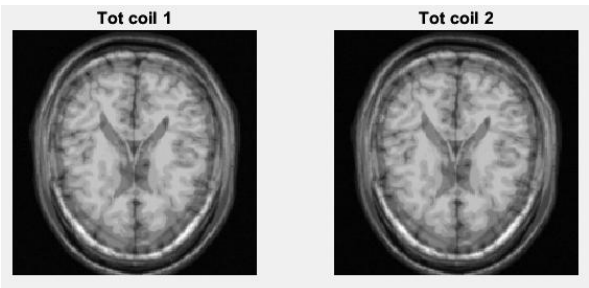
Initially, we compared the results of the SENSE and CAIPIRINHA methods for two scenarios: uniform (left) and exponential (right) coil sensitivities within the slices. In both scenarios, the slice modulation parameter was set to 0.95 to simulate a realistic situation where the two slices are relatively close together, resulting in minimal differences between the coil sensitivities of the two slices.

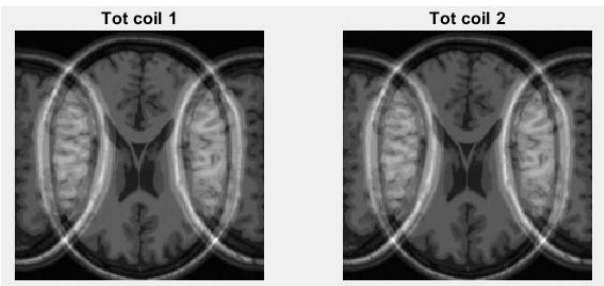




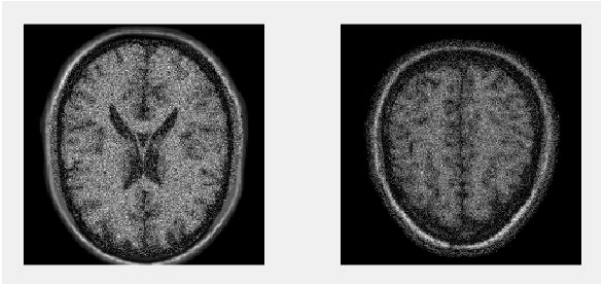
Uniform coil sensitivities: results for SENSE (left) vs CAIPIRINHA (right):

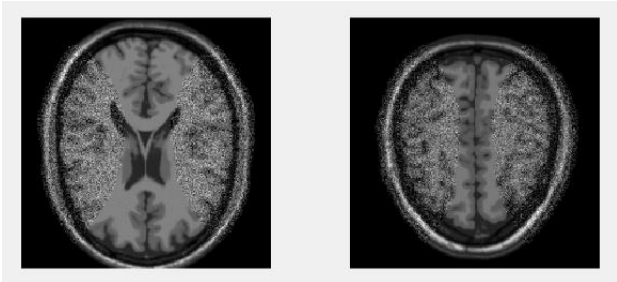
Images seen by the coils (before reconstruction): in the case of SENSE there's no phase shift between the two slices and they totally overlap in the coil images; in the case of CAIPIRINHA the second slice is effectively shifted by half of the FOV in the horizontal direction.



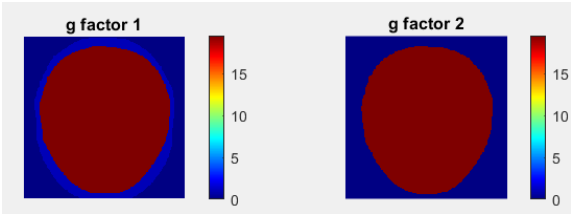


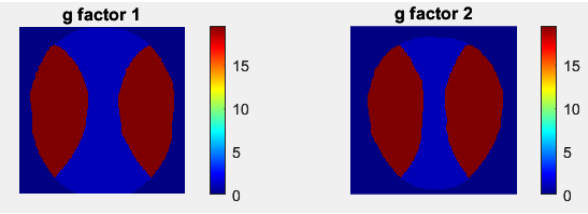
Reconstructed images: in the case of SENSE, the reconstructed images exhibit noise uniformly across the entire slice area, while in the case of CAIPIRINHA they display noise just in the overlapping regions of the coil images, resulting in better overall quality.





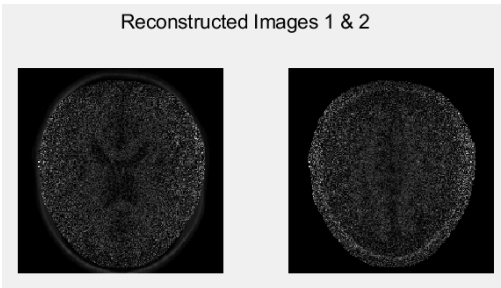
g-factor: SENSE and CAIPIRINHA have the same value of g-factor in the overlapping regions, but in the case of CAIPIRINHA we have the advantage that the overlap area is smaller. (Note: the g-factor in the background results equal to 0 because it's not calculated there, due to masking).

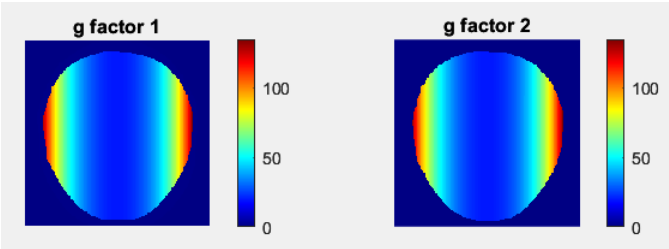




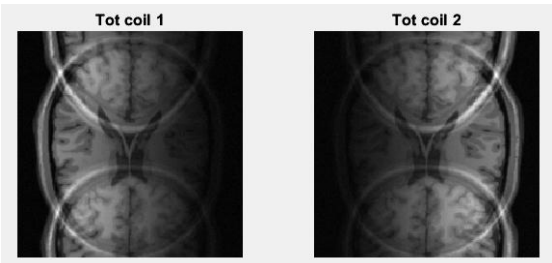
Results for exponential coil sensitivities

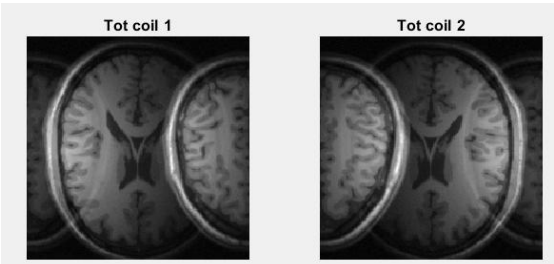
SENSE: the reconstructed images are very noisy and it's not possible to recognize the features of the original slices. The poor performance aligns with the high g-factor, which has an average value of 38.2 within the slice region. In this case, the g-factor is not uniform across the slice, unlike in the case of uniform coil sensitivities; this variability arises from the spatially dependent coil sensitivities.

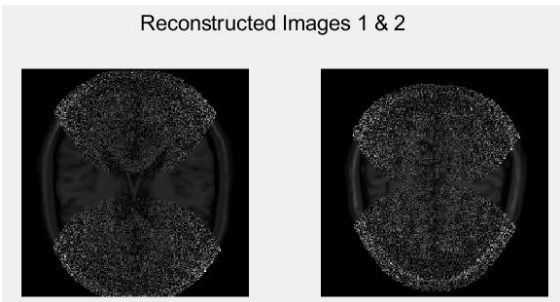


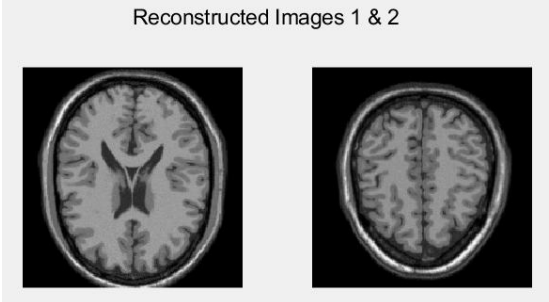


CAIPIRINHA: comparison between vertical (left) and horizontal (right) shift.

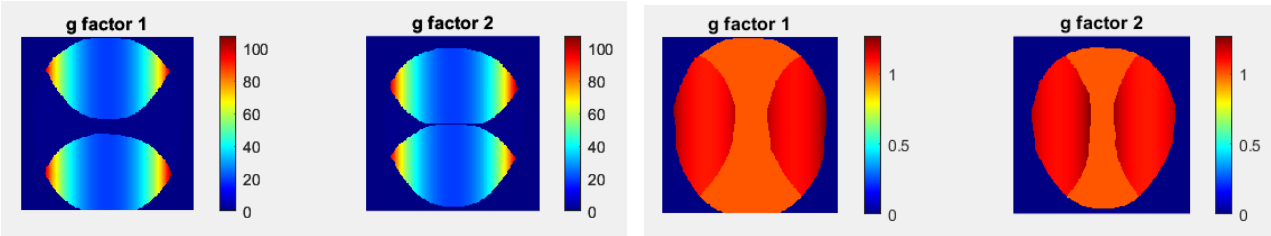








Better performance was observed with a horizontal shift compared to a vertical shift. This improvement may be attributed to the fact that, in the case of a vertical shift, the overlapping regions of the two slices exhibit very similar coil sensitivities. In contrast, with a horizontal shift, the overlapping regions have distinct coil sensitivities, making it easier to separate the slices and significantly improving the reconstruction quality. This observation is further supported by the **g-factor** values: with a vertical shift, the g-factor reaches approximately 20 in the central region and becomes extremely high near the boundaries. In contrast, with a horizontal shift, the g-factor is considerably lower, remaining at 1 in the non-overlapping regions and around 1.1–1.2 in the overlapping regions.



In conclusion, under the conditions of exponential coil sensitivities—representing a more realistic scenario—the advantages of the CAIPIRINHA method over SENSE are more clearly demonstrated.

Results of SENSE & CAIPIRINHA with varying slice modulation:

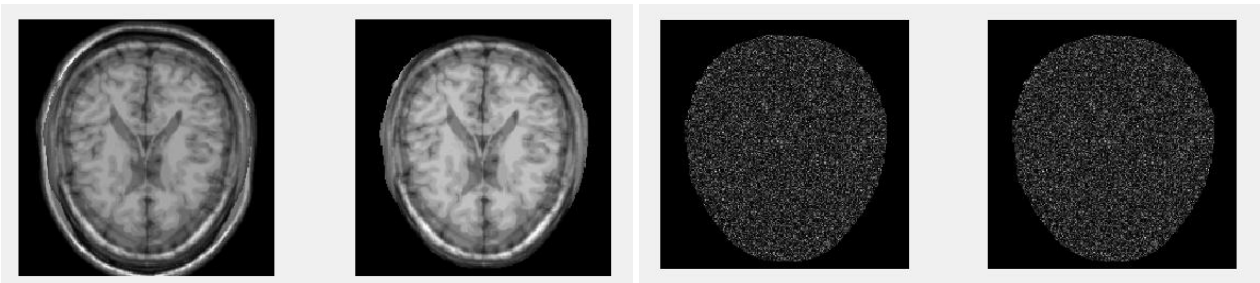
Secondly, we analysed the impact of varying levels of coil sensitivity differences between the two slices, specifically examining the extreme cases where the slice modulation approaches 1 and 0.

Worst case: slice_modulation approaching 1 (identical coil sensitivities in the two slices)

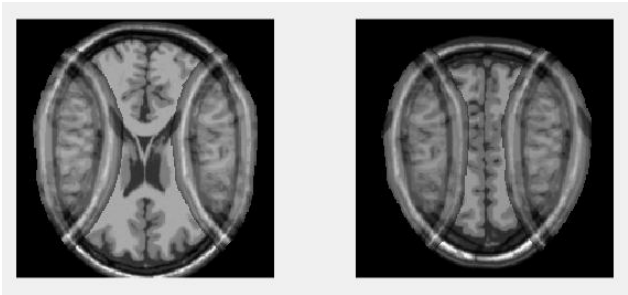
Uniform coil sensitivities: in this scenario, the reconstruction failed for both SENSE and CAIPIRINHA. Specifically, when the slice_modulation parameter was exactly 1, the reconstructed images for both methods consisted of the two slices superimposed. With slice_modulation set to 0.9999, the reconstructed images became entirely noisy in the overlapping regions, making it impossible to discern any details of the slices.

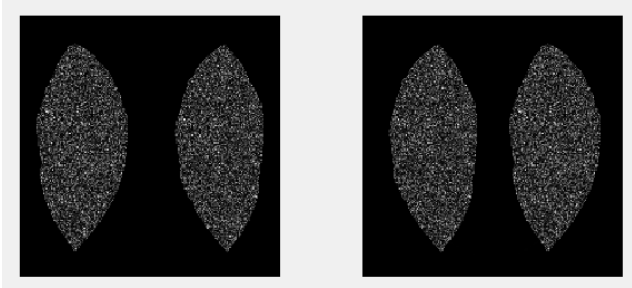
Reconstructed images for slice_modulation = 1 (left) and slice_modulation = 0.9999 (right):

SENSE:



CAIPIRINHA:

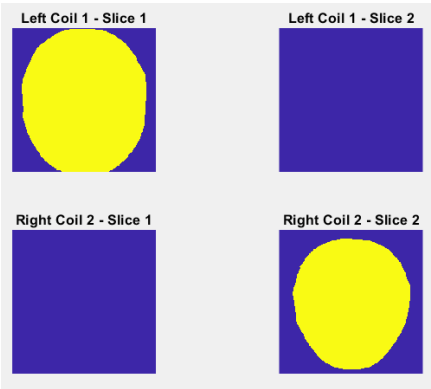


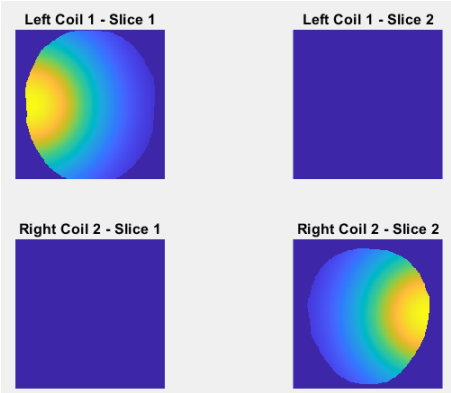


Exponential coil sensitivities: in this case, SENSE fails completely, producing completely black images when slice_modulation is set to 1, and highly noisy images - similar to those observed with uniform coil sensitivities - when slice_modulation is 0.9999. In contrast, CAIPIRINHA delivers a perfect reconstruction. This is another demonstration of how exponential coil sensitivities are better compared than uniform coil sensitivities to evaluate the performances of these methods.

Ideal case: slice_modulation = 0:

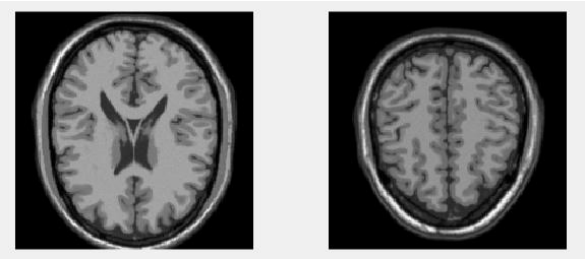
In this case, coil 1 was sensitive exclusively to slice 1, while coil 2 was sensitive exclusively to slice 2.

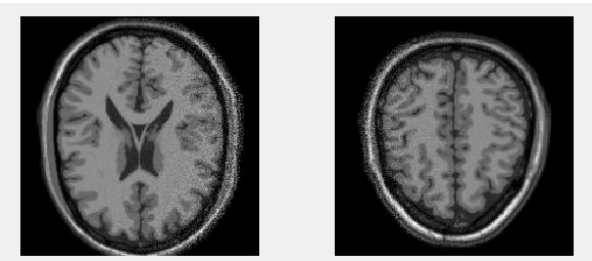




Results: SENSE and CAIPI perform identically, delivering very good reconstructions with a consistent g-factor of 1.

Reconstructions for uniform (left) and exponential (right) coil sensitivities (for both SENSE and CAIPI):





In particular, for uniform coil sensitivities, the reconstruction is perfect. However, with exponential coil sensitivities, slight noise is observed on the right side of slice 1 and the left side of slice 2. This phenomenon is analysed in the next section.

g-factor: the following table summarizes the average g-factor values within the regions of the slices, calculated for the limit cases of slice_modulation discussed earlier. These values are consistent with the results obtained.

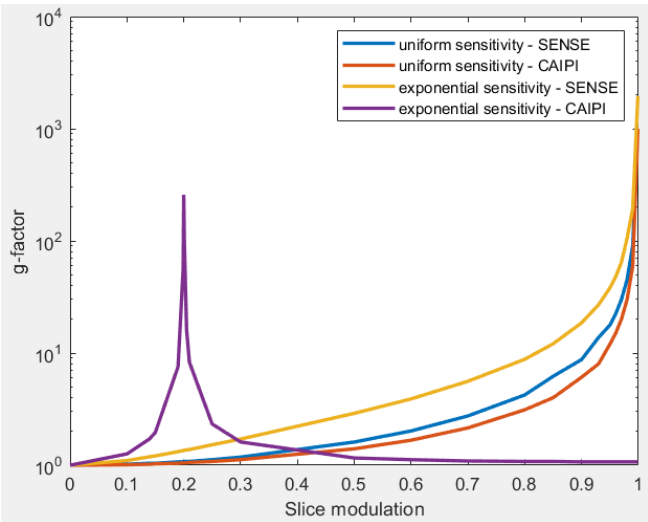
g-factor:	SENSE (uni)	CAIPI (uni)	SENSE (exp)	CAIPI (exp)
Slice modulation = 0.9999	9093.25 ✗	5965.3 ✗	19570 ✗	1.07 ✓
Slice modulation = 0	1 ✓	1 ✓	1 ✓	1 ✓

Note: In the case of uniform coil sensitivities, the g-factor value in the overlapping regions is identical for both SENSE and CAIPI. However, the average g-factor in the slice regions is lower for CAIPI, as the overlapping area is smaller in this case.

Considerations about the g-factor:

The analysis of g-factor values was extended across the entire range of slice_modulation values, from 0 to 1, for both SENSE and CAIPIRINHA, considering both uniform and exponential coil sensitivity.

Plot of the average value of the g-factor in the slice regions:

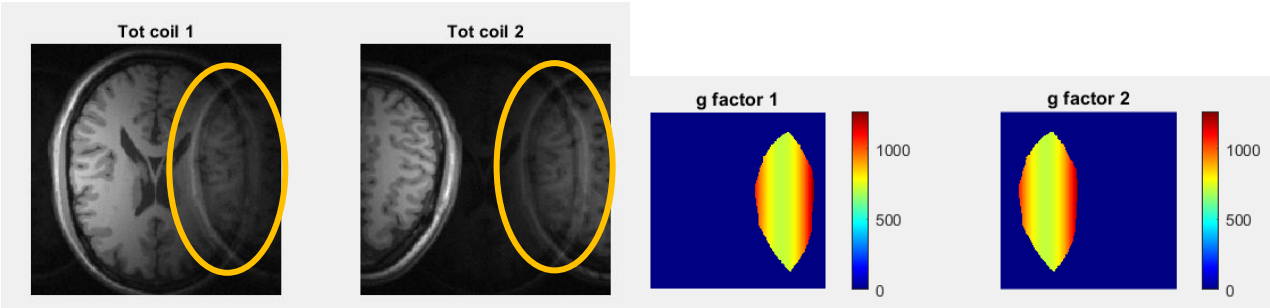


Uniform coil sensitivities: the g-factor is identical for both SENSE and CAIPI in the overlapping regions, meaning there is no apparent advantage of CAIPI over SENSE in this case. When computing the average g-factor within the slice regions, CAIPI shows slightly lower values compared to SENSE, due to the reduced overlapping area. Nonetheless, the performance of both methods remains quite similar.

Exponential coil sensitivities:

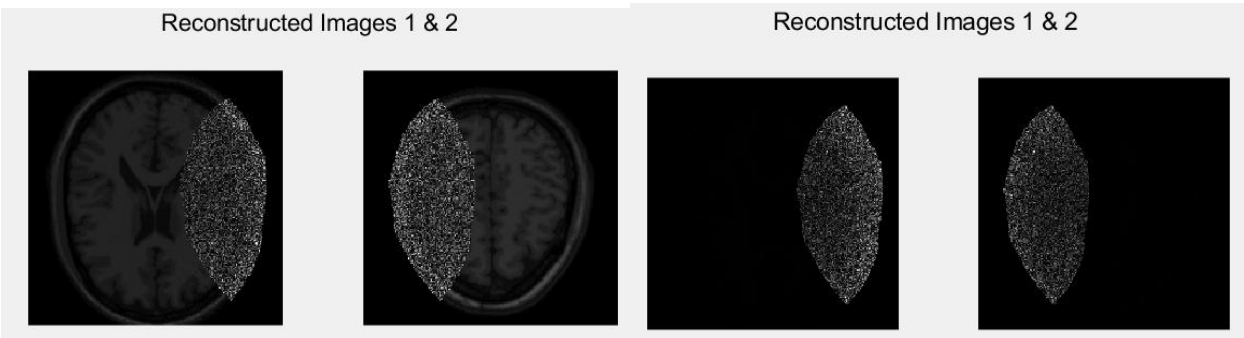
- For high values of slice modulation, the g-factor for CAIPI remains nearly constant and close to 1 (around 1.07–1.08), resulting in significantly better performance for CAIPI compared to SENSE.
- For low values of slice modulation, a unique behavior is observed for CAIPI's g-factor: it increases as slice modulation decreases, reaching a peak at slice_modulation = 0.20. Then, as slice_modulation approaches 0, the g-factor decreases again to its minimum value of 1.

Case slice modulation = 0.2:



This phenomenon may be attributed to the coil sensitivities of the overlapping regions of the slices becoming roughly equal for very low slice_modulation values. For instance, with coil 1, the sensitivity of the left side of slice 2 (which is reduced due to the slice_modulation factor) becomes equal to the sensitivity of the right side of slice 1 (also reduced due to distance from the coil). As a result, the reconstruction fails in this case.

Below we can see the results for slice_modulation = 0.21 (left) and slice_modulation = 0.2 (right):



This issue does not occur with SENSE using exponential coil sensitivities, or with either SENSE or CAIPI using uniform coil sensitivities, where the g-factor continues to decrease as slice_modulation decreases.

Conclusion: Overall, CAIPI with exponential coil sensitivities offers the best performance (g-factor), except in this specific scenario. In this case, it is preferable to use the SENSE algorithm, as a perfectly uniform coil sensitivity is not practically achievable.